SOFTWARE TESTING REPORT

End-to-End E-commerce Workflow Testing using Selenium WebDriver and Python

Student Name	Bhavesh Patil
Enrollment Number	2504030100419
Programme	Bachelor of Computer Applications (BCA)
Semester	3rd Semester
Website Under Test	SauceDemo — saucedemo.com
Testing Tool	Selenium WebDriver with Python

Prepared for academic practical submission

1. Executive Summary

This report documents an automated end-to-end test of the SauceDemo e-commerce website. The Python script uses Selenium WebDriver to open the site, sign in with a test account, add a product to the cart, navigate to checkout, enter checkout details, verify the next checkout page, and exercise browser back/forward navigation.

The captured run output reports PASS for login, product selection, cart navigation, multi-page workflow, and back/forward navigation. The run ended with exit code 0. The screenshots are included as visual evidence; the blank/loading browser screens are recorded as intermediate states rather than test failures.

2. Aim

To automate and validate a multi-step e-commerce user journey using Selenium WebDriver and Python.

3. Objectives

• Launch Chrome through Selenium and open SauceDemo.
• Log in using the provided test credentials.
• Confirm that the inventory/products page is reached.
• Add the Sauce Labs Backpack to the cart.
• Open the cart and proceed to checkout.
• Enter first name, last name, and postal code.
• Verify the checkout overview page and test back/forward navigation.

4. Tools and Environment

Component	Details
Language	Python
Automation library	Selenium WebDriver 4.x
Browser	Google Chrome
Test site	https://www.saucedemo.com/
Wait strategy	WebDriverWait with explicit expected conditions (30 seconds in the latest version)

5. Test Methodology

The test uses explicit waits to allow elements and page transitions to become ready before interaction. Assertions validate key URL states. A try/finally block closes the browser even if an exception occurs.

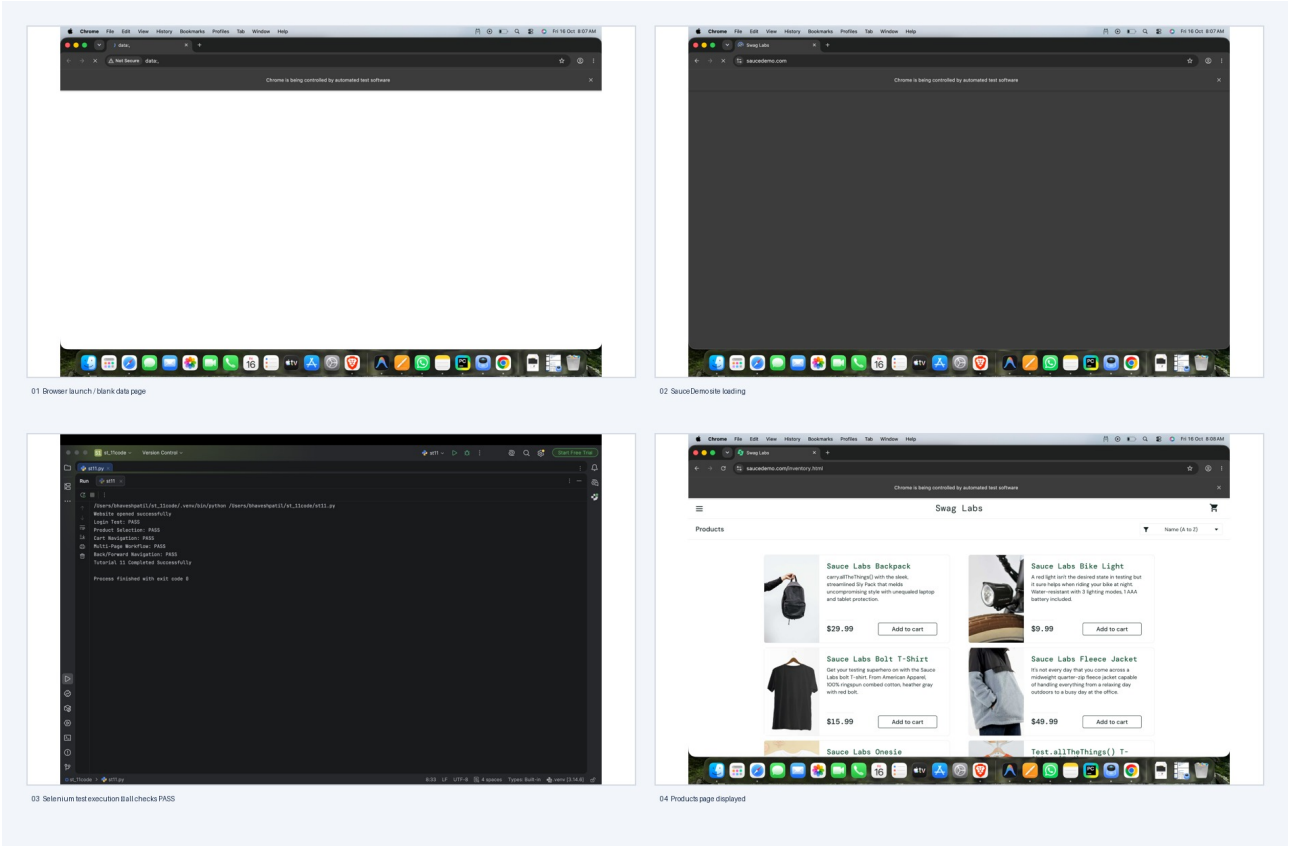
6. Test Cases and Results

ID	Test scenario	Expected result	Observed result
TC-01	Open SauceDemo	Website opens	Opened successfully (console)
TC-02	Login with standard_user	Inventory page appears	PASS
TC-03	Add Sauce Labs Backpack	Product added to cart	PASS
TC-04	Open shopping cart	Cart page appears	PASS
TC-05	Open checkout	Checkout form appears	Workflow proceeded
TC-06	Submit checkout information	Checkout overview appears	PASS
TC-07	Browser back and forward	Navigation commands execute	PASS

Note: The script verifies that the checkout-step-two URL is reached; it does not place the order or validate an order-confirmation page.

7. Screenshot Evidence

The collage below contains the four screenshots supplied for this run. The execution console shows the automated checks passing, and the final browser screenshot shows the SauceDemo inventory page.



8. Detailed Visual Evidence

Figure 1 — Browser launch / intermediate data page

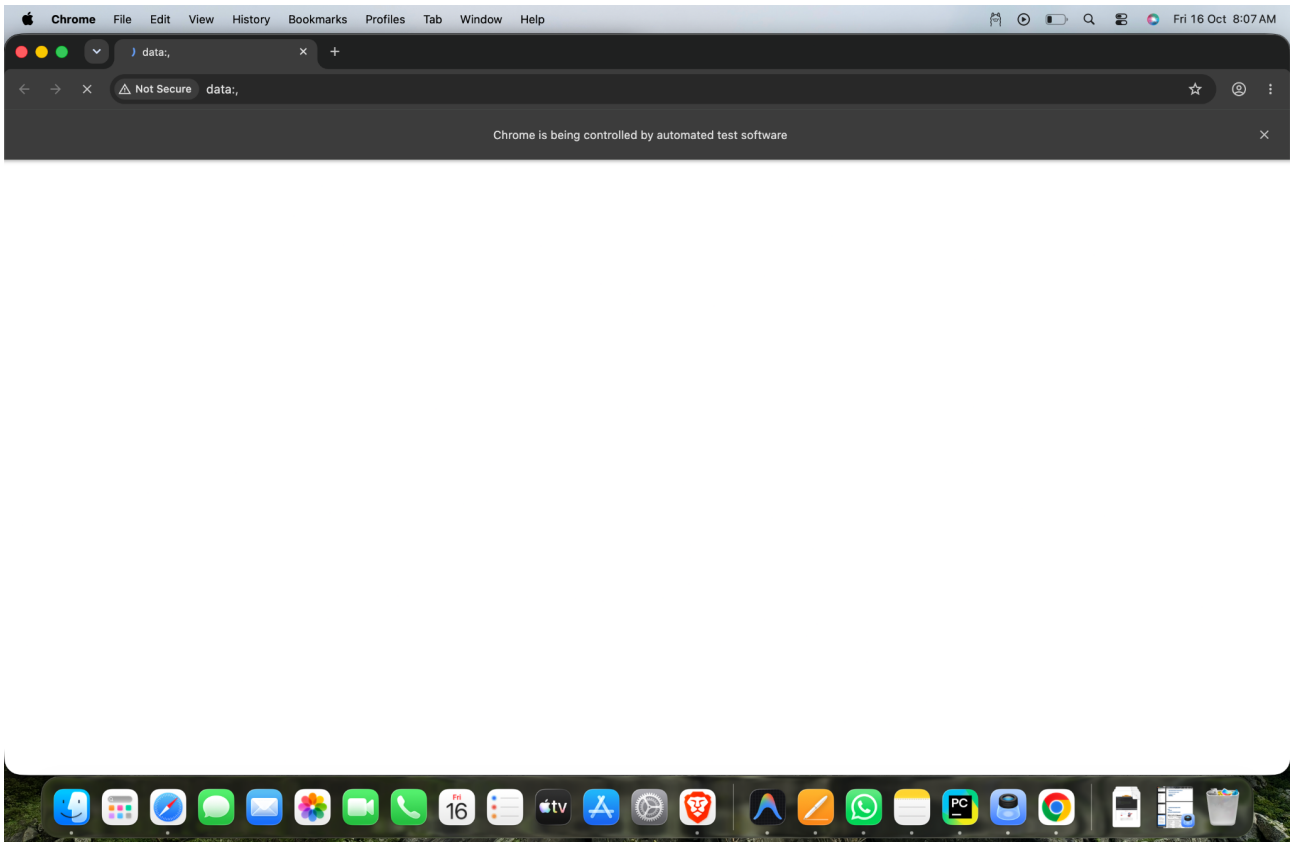


Figure 2 — SauceDemo website loading

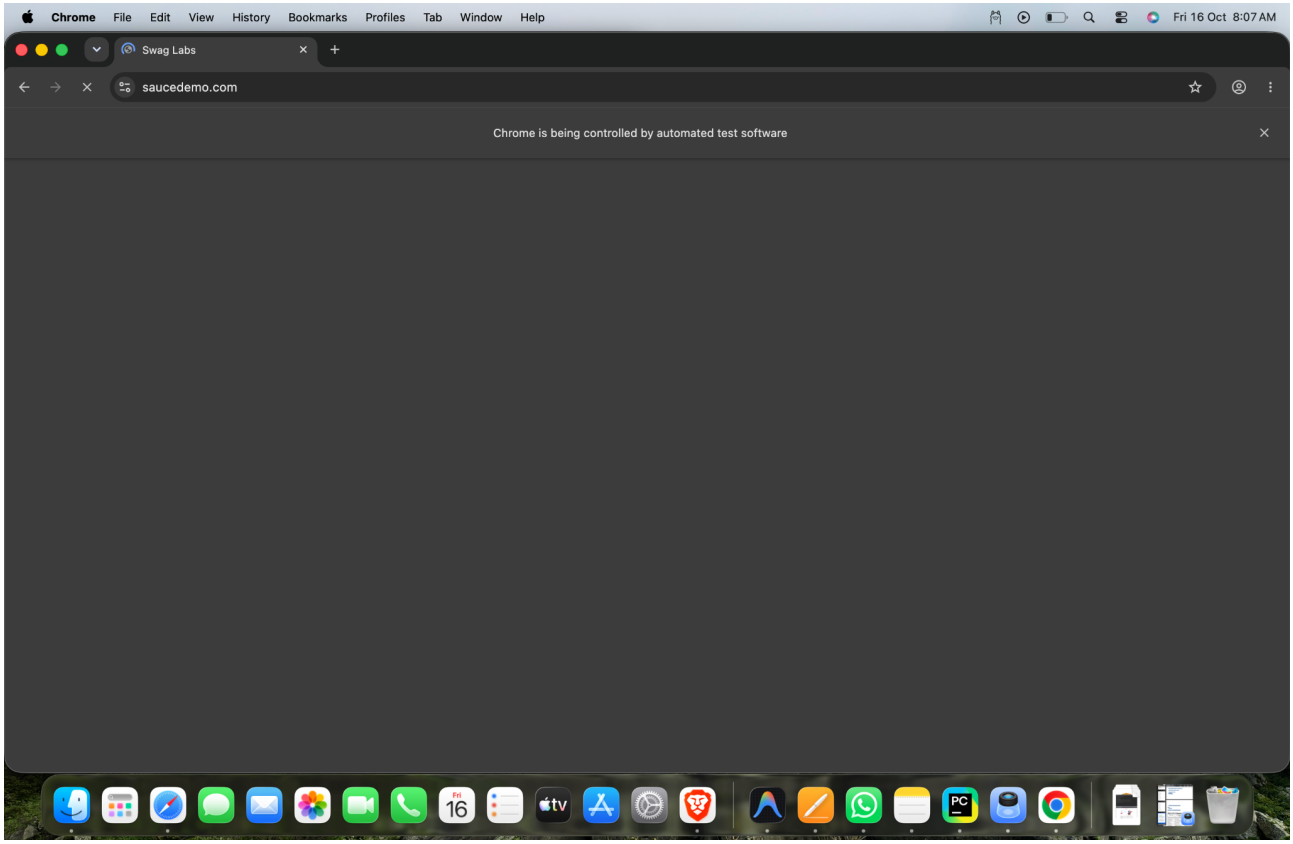


Figure 3 — Test execution console showing PASS results

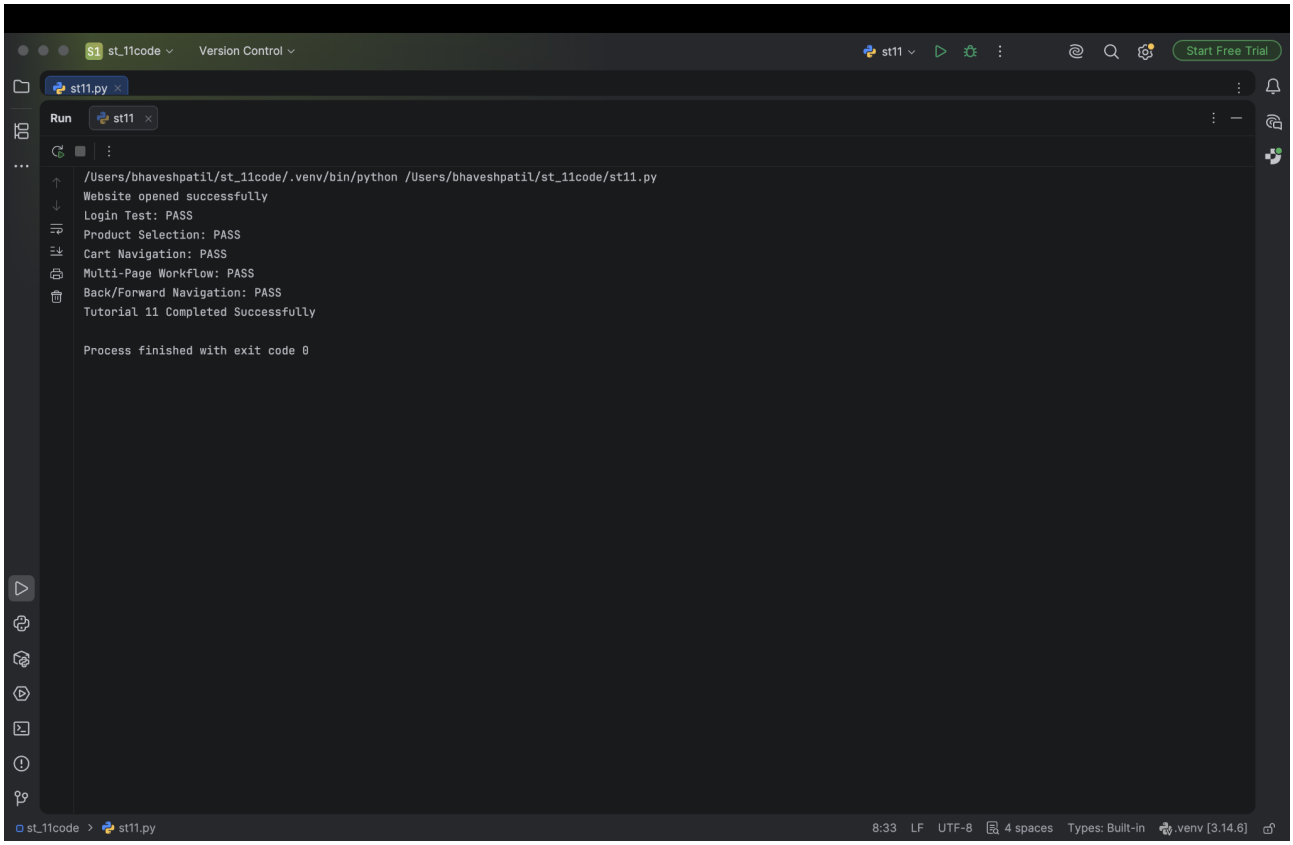
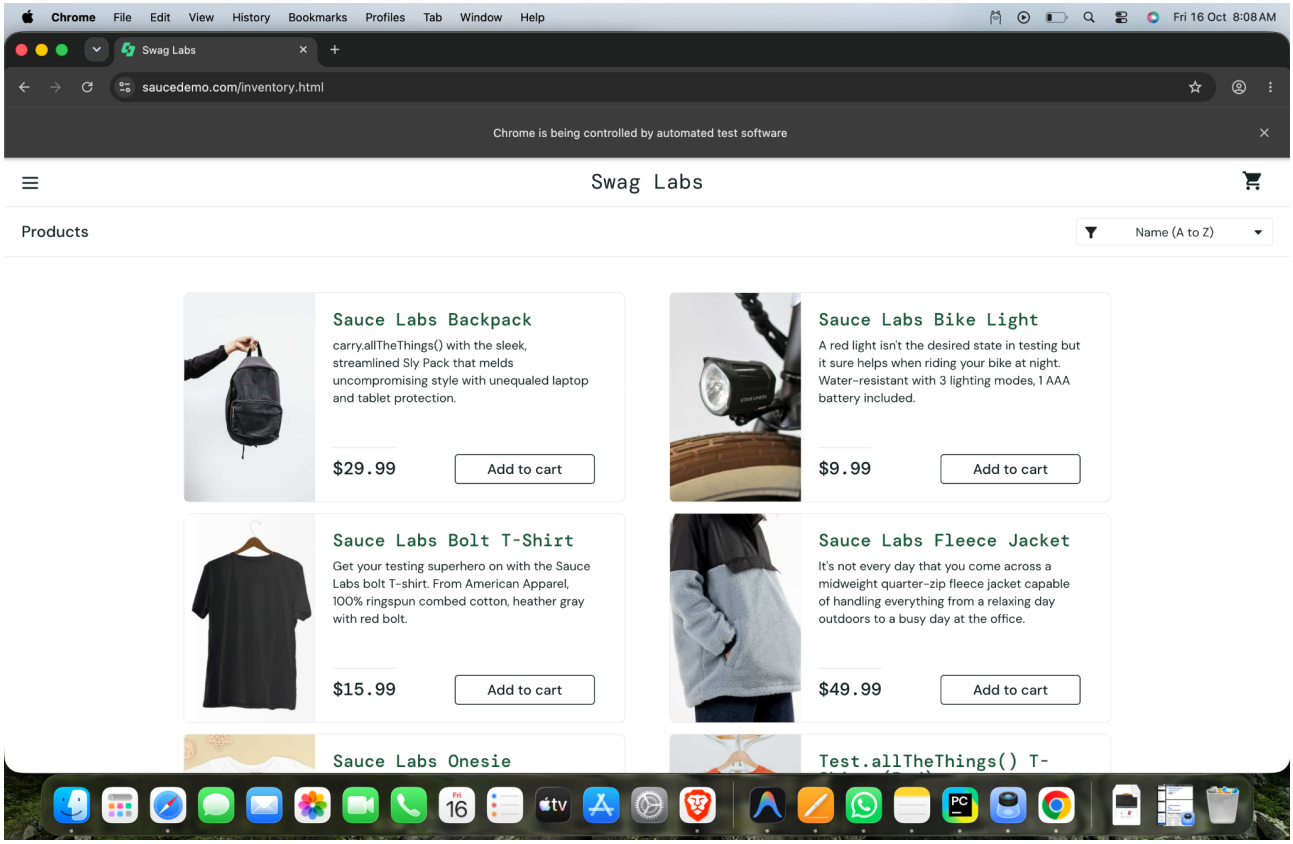


Figure 4 — SauceDemo inventory/products page



9. Conclusion

The Selenium automation completed the main SauceDemo shopping workflow successfully according to the console output. Login, product selection, cart navigation, checkout progression, and browser navigation checks passed. The test demonstrates how explicit waits and assertions can improve the reliability of UI automation.

10. Recommendations

- Add a check that the cart badge shows one item after adding the backpack.
- Verify checkout form validation using missing or invalid fields.
- Add a final order-placement test and assert the confirmation message.
- Capture screenshots automatically when a test fails.
- Use a test framework such as pytest or unittest for structured reporting.